Round 4: follow-up experiments on items 1–3 (Addendum H)

Status. Nine follow-up experiments (H1–H9) were frozen in `PROTOCOL.md` (Addendum H, `f3bcf9a63`) before any of them ran. Amendments H-a and H-b were committed before the runs they govern. Seven experiments ran and are recorded:

- H1, H2 (with H3 folded in) and H6 on RunPod CPU pods;
- H4, H5 and H7 in this session's container.

H8 was not run, and H9 is answered by existing studies. Every cost is a counted operation. Every discrete log was verified (`k·P = Q`): 3,860 in total.

Where it ran. The repo's `ephemeral` runner could not authenticate (the container has an AWS access-key ID but no secret). Jobs went instead through `runpod/podrun.py`. A pod fetches the tool at a pinned, pushed commit from the public repository, checks each file's SHA-256, runs the command fixed at launch, and serves only read-only results. Nothing is uploaded into the pod. Every pod was terminated after its fetch, and each run directory keeps `pod.json`, `host.txt` and `rusage.txt`.

Answer

Item 3: `n = 2` through genus-2 Jacobians.

1. **The route reaches prime-order curves [E].** Any `E/F_{p²}` with `j ∉ F_p` is glued to `E^σ` along `ψ(a) = a^{p³}` on the 2-torsion roots. This gives a genus-2 `C/F_p` with `Jac(C) ~ Res E` and an explicit transfer.
   - It descends whenever the 2-torsion cubic is irreducible or split over `F_{p²}`.
   - With exactly one rational 2-torsion point, no Galois-equivariant gluing satisfies the descent cocycle. So `h ≡ 2 (mod 4)` is not reached, which matches `4 | #E` throughout the Scholten family.
   - Checks G1–G5 hold on every instance (see the checks table).
2. **On prime-order `E` (`h = 1`)** the first-cycle IC needs **0.43–0.44·p** Jacobian operations, against rho's 0.78–0.96·p: about half.
3. **Per-operation cost.** The reduced sextic-model formulas cost 41 M + 1 I per addition and 51 M + 1 I per doubling, against 12 M + 1 I for an `E(F_{p²})` addition.
4. **Ranking at `h = 1`.** With the split test priced honestly (one Legendre symbol at one `I`), the pricing band is **0.59–1.94 (unresolved) at every size from 28 to 40 bits**, with the IC cheaper in up to 6 of 12 cells.
   - Distance to rho at measured cost (`I = 3M`): 1.43–1.81×.
   - Break-even: a genus-2 addition of about 22 M + I. That is the best published figure, not yet measured here.
5. **The cofactor law (H1) closes the large-h hope.**
   - Arm J's trials scale as `h^{−0.402}` (95% [0.380, 0.424]); rho's as `h^{−0.468}`, near the theoretical ½.
   - Arm J's count advantage is largest at small `h` and is gone by `h ≈ 16`.
   - The frozen decision rule applies: on large-cofactor curves (FourQ-like `h = 392`) this route never beats rho on count.
6. **Larger sizes (H6, 44–48 bits)** reproduce the constants.
   - Rho: `steps/p` 0.33 and 0.45, against theory 0.31 and 0.44.
   - Arm J: `steps/p` 0.30 and 0.31, band unresolved.
   - Arm D (decomposition): 28–87× rho.
7. **Memory (H4) is the real limit.** Time × memory does not stay near `p²/9`, as predicted: it falls from 0.040·p² to 0.004·p² as θ goes from ½ to ⅛. With few stored relations, the walk solves mostly through repeated relations, which is distinguished-point rho inside the IC walk. A memory-capped Arm J turns back into rho, so the IC's constant-factor count advantage exists only with Θ(p) memory.
8. **Automorphisms (H5) [E].** An exact census over `p = 5…13` finds 6, 6, 14 and 14 classes that are simple over `F_p`, Res-shaped, and have automorphisms beyond `±1` (orders 4 to 120). The total mass equals `p³` exactly at every `p`. These are candidates for an IC gain of `k` against rho's √k. Measuring that needs such curves at cryptographic size and a rho that uses the same automorphism; not done.

Item 1: decomposition over `F_{q^n}`.

9. **Coset confinement measured on the target subgroup (H7) [E].** This is coverage of the ℓ-subgroup itself, not the whole group.
   - **Binary `n = 2`:** subfield over null gives a ratio of **1.86 ± 0.16** (21 curves). That confirms the corrected reading of round 3: the subfield factor base reaches 0.75 of `2E` and of each deeper `2^k E`, against a Poisson value of 0.39.
   - **Odd characteristic, size-matched split twists (`E ≅ Y² = Π(X − d e_i)`, `e_i ∈ F_q`, `d` a square), `n = 2`:** ratio **2.39 ± 0.21** (12 curves). The subfield factor base covers the whole ℓ-subgroup (1.0, against 0.27–0.71 for null).
   - **Generic odd curves:** 1.17 ± 0.10, near the no-effect value.
   - **`n = 4` cells:** too small to read (s.e. 0.13–0.71).
   - The gain is bounded by the cofactor's 2-part. It is a cost-model fact about these curve families, not a new algorithm, and prime-order curves gain nothing.

Item 2: prime fields.

10. **H9 needs no new run.** The PKM solving degree is the subject of `research/notes/index-calculus/RESEARCH_PKM_TOWER_ORACLE.md`, rounds 1–7 and later studies (for example `D = 7` at `m = 3`, `N = 18`, and at `m = 4`, `N = 16`). Re-measuring would duplicate them; this round cites them. Round 3 settled coverage: no effect.

Not run: H8 (decomposition crossover at `n = 3`). The repo's Semaev tools are binary-field only. A descended-`S_4` solver over `F_{p³}` (Gröbner or resultant) is a separate build. It is the next item-1 experiment.

H1: the cofactor law (run `H1-hsweep-run-20261018T2240Z`, RunPod, 16 vCPU)

Scale. The Scholten family, with `h ∈ {4, 8, 12, 16, 24, 32, 48, 64, 96, 128}` × `log2 #E ∈ {32, 36, 40}`. Each cell has 4 instances × 10 targets, with arms R and J, plus Jind at 32 bits. All answers verified.

Fits. On cell means, with a 2,000-sample bootstrap over instances:

arm	α (trials/p ∝ h^−α)	95% interval
J (walk)	0.402	[0.380, 0.424]
Jind (independent trials, 32 bits)	0.368	[0.334, 0.400]
R (rho)	0.468	[0.446, 0.490]

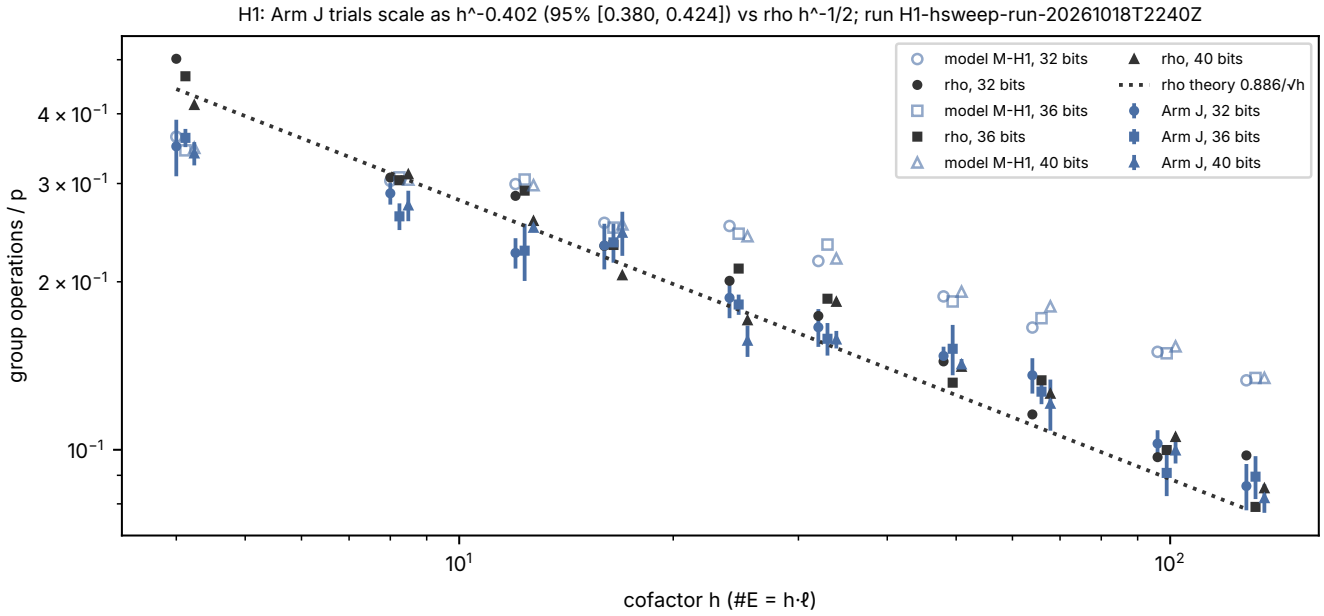


Figure 1: H1

Model M-H1. This is the uniform admissible-pair graph on the instance's real degree-1 points and their cofactor classes, with sign parity. It matches the measured trials within about 10% at `h ≤ 16`. At `h = 64–128` it overpredicts by 1.3–1.6×. Its uniform-pair assumption does not hold there: relations are not uniform over admissible pairs once the classes are small. The measured law is the result; the model is diagnostic only.

H2/H3: the general gluing (run `H2-glue-run-20261018T2240Z`, RunPod, 16 vCPU)

Construction.

- Solve `R ab + S a + T b = 1` over the étale algebra `F_{p²}[z]/(g)`, with `b = ψ(a) = z^{p³}`. This gives `s_2 = −T/R`, `t_2 = −S/R`, `k = s_2t_2 + 1/R` and `s_1 = k·g(s_2)`.
- That defines `C: y² = g(s_1x² + s_2)/s_1²` over `F_{p²}`.
- Descent datum: `x ↦ λ/x^σ, λ² = k/N(s_1)`.
- The `F_p` model comes from a Baer-subline cross-ratio map. A rational point is moved to infinity, so the model has two rational points at infinity. It is twisted when needed, so that `τ` is nonzero on the `ℓ`-part.
- The IC columns are `x_P = [2P − D∞]`, so a relation reads `(x_P + x_Q)/2`.

Checks. On 16 instances plus G5:

check	result
`t_2 = σ(s_2), λ² ∈ F_p`	16/16
G1: `#C(F_p) = p + 1; #C(F_{p²}) = p² + 1 − 2t` (`p < 400`)	16/16; 8/8
G2: associativity, inverse, `N·A = 0`	all true; `N·A` "degenerate" twice at `p ≤ 277`, never false
G3: homomorphism, `τ ≠ 0, ℓ·τ = 0`	all true; one "degenerate" at `p = 257`
G5: reduced formulas = generic law	2,400/2,400 pairs

Measured (`log2 #E` 28–40; 4 instances × 10 targets per cell; all answers verified). Group operations per `p`, mean ± s.e.:

bits	h	Arm J	rho	Arm J cost / rho, pricing band	verdict
28	1	0.435 ± 0.017	0.956 ± 0.064	0.59–1.51	unresolved
32	1	0.426 ± 0.018	0.844 ± 0.069	0.71–1.71	unresolved
36	1	0.439 ± 0.014	0.778 ± 0.076	0.82–1.94	unresolved
40	1	0.443 ± 0.016	0.860 ± 0.068	0.76–1.80	unresolved
36	4	0.337 ± 0.018	0.403 ± 0.041	1.18–2.79	rho cheaper
40	4	0.350 ± 0.018	0.445 ± 0.031	1.14–2.69	rho cheaper

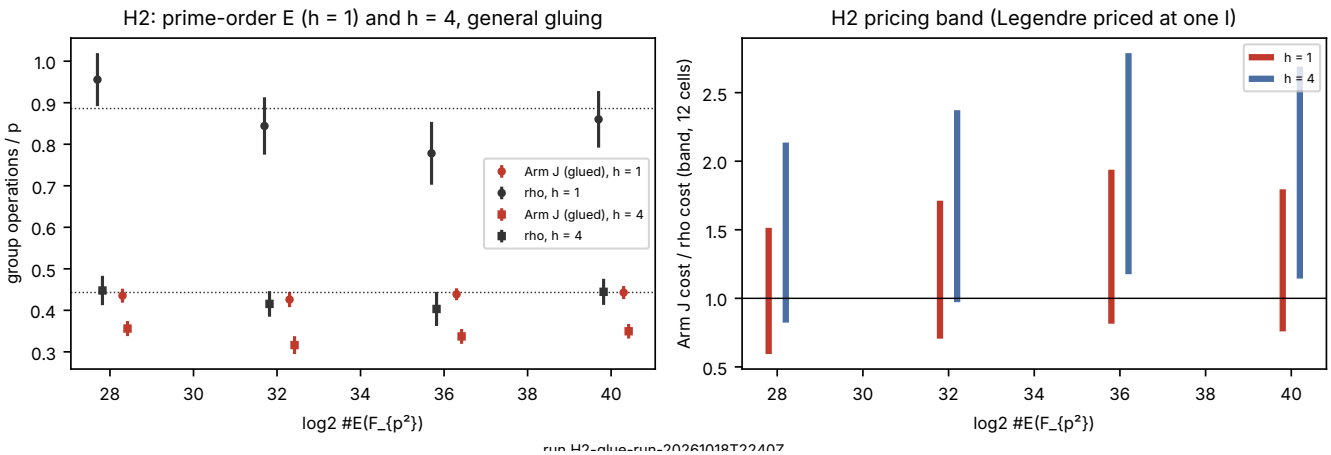


Figure 2: H2

Distance to rho (prime-order `E/F_{p²}`). About 1.8× at 40 bits under measured costs (1.78× with the H3 Legendre split test at one `I`, as the band prices it; the 1.68× in `glue_summary.txt` leaves the split test out, corrected 2026-10-24 on the validator's report). Under the IC-favourable price (one Jacobian operation = one `E(F_{p²})` addition), the IC is cheaper: 0.76× at 40 bits. Closing the gap needs a genus-2 sextic-model addition of about 22 M + I, measured. This round's reduced formulas take 41 M + I.

H4: memory against time (run `H4-memsweep-run-20261018T2209Z`)

bits	θ	trials/p	stored/p	time·memory/p²
32	½	0.563	0.0719	0.0405
32	¼	0.624	0.0201	0.0125
32	⅛	0.700	0.0056	0.0039
36	½	0.498	0.0624	0.0311
36	¼	0.588	0.0182	0.0107
36	⅛	0.617	0.0048	0.0030

At θ = 1 (`h = 4`) the trials are about 0.35·p, with about 0.18·p stored.

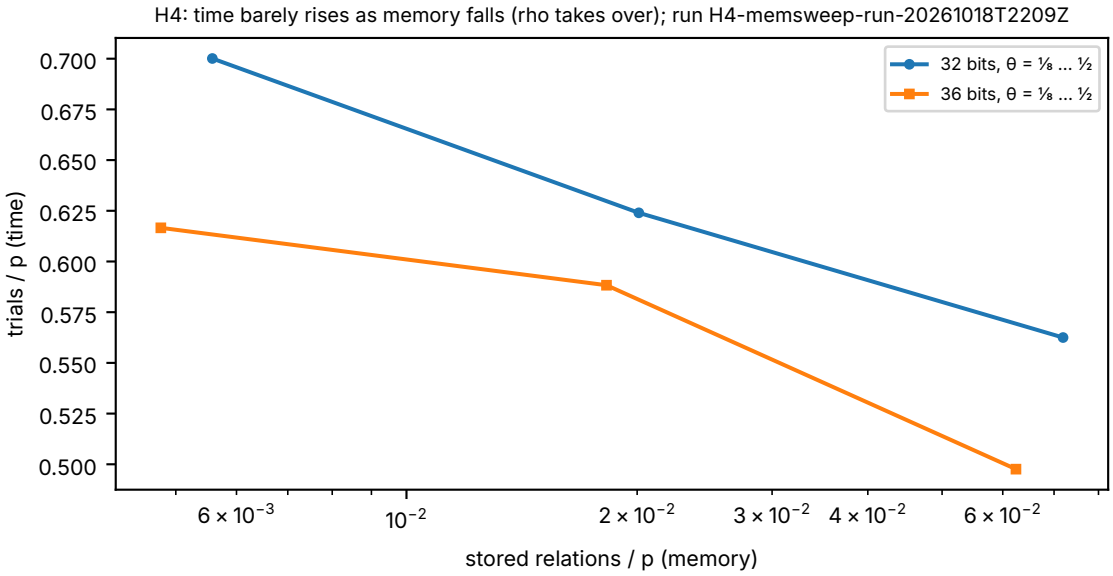


Figure 3: H4

The prediction (time ∝ 1/θ) is falsified. As θ falls, the answers come from relations the walk repeats, so the arm becomes rho with distinguished points (H4 used the walk arm; a Jind-θ arm would isolate the graph mechanism, and was not run). The IC's count advantage at θ = 1 depends on Θ(p) relation storage. At cryptographic sizes that is the binding constraint, and the comparison should be read as area-time.

H5: automorphisms (run `H5-aut-census-run-20261018T2227Z`) [E]

Classes simple over `F_p` with even L-polynomial (`a_1 = 0`), by `|Aut|`, which includes the hyperelliptic involution:

p	mass Σ1/|Aut| (= p³)	|Aut| = 2	4	6	8	12	24	120	with |Aut| > 2
5	125	31	1	4	–	–	–	1	6

7	343	85	3	2	–	–	1	–	6
11	1331	306	3	9	1	1	–	–	14
13	2197	450	6	7	–	–	1	–	14

So Res-shaped, simple Jacobians with extra automorphisms over `F_p` exist. Whether the IC's gain of `k` beats rho's √k on them needs the same automorphism granted to rho in `Jac(C)`, and needs a family at large `p`. Not measured.

H6: larger sizes (run `H6-large-run-20261018T2210Z`, RunPod, 16 vCPU)

Arms R, J and D at `c = 2`; 3 instances × 10 targets per size; all 180 answers verified.

bits	rho steps/p	theory	Arm J steps/p	Arm J band	Arm D band
44	0.328 ± 0.026	0.313	0.295	0.99–2.96, unresolved	28.4–80.9, rho cheaper
48	0.448 ± 0.042	0.443	0.311	0.77–2.32, unresolved	28.1–86.8, rho cheaper

H7: coverage of the target subgroup (run `H7-subgroup-run-20261018T2209Z`) [E]

Ratio of ℓ-subgroup coverage, subfield over null; mean ± s.e. over curves:

cell	n = 2	n = 4
binary (`m = 4…10` / `3…5`)	1.86 ± 0.16 (21)	2.07 ± 0.71 (9)
odd generic	1.17 ± 0.10 (12)	1.33 ± 0.53 (9)
odd split twist, frozen (`d` generic)	1.18 ± 0.55 (12)	0.40 ± 0.15 (9)
odd split twist, size-matched (`d` square)	**2.39 ± 0.21** (12)	0.74 ± 0.13 (9)



Figure 4: H7

Why the frozen split-twist cell is noisy. With `d` a non-square, the factor base is empty apart from the 2-torsion points (Amendment H-a).

Why the `n = 4` cells are unreadable. Their ℓ-subgroups are small (tens to hundreds of points).

Mechanism.

- **Binary:** `Tr(x) = 0` for `x ∈ F_q`.
- **Odd split twists:** every `X − d e_i ∈ d·F_q` has the same quadratic character.

In both cases the factor base sits in one coset of `2E`, and its sums land in `2E`, which contains every target.

Checks

check	result
every discrete log verified (`k·P = Q`)	H1 2,800; H2 640; H4 240; H6 180 (3,860 in this round, all verified)
rho positive control (K6) at 44/48 bits	matches `0.886·√ℓ/p` within 1 s.e.
G1–G5 (gluing)	see H2; no false check
H5 mass `Σ 1/|Aut| = p³`	exact at p = 5, 7, 11, 13
unit tests	n2_routes 5, decomposition_coverage 5, genus2_aut_census 1: pass

What is not established

- **H2.** The 22 M + I break-even figure is an estimate. No formula at that cost was implemented here.
- **H4.** The memory result uses the walk arm.

- **H5.** No measurement beyond the census.
- **H7.** The `n = 4` cells are underpowered.
- **H8** was not run.
- **Novelty.** None of these results has a novelty audit. The general gluing is very likely Scholten's (2003) construction, written down here with the `a^{p³}` cocycle made explicit. The statement that `h ≡ 2 (mod 4)` is unreachable needs checking against that paper.

Next improvements

1. **Measure a cheaper sextic-model addition.** This decides `h = 1`; the target is 22 M + I or less. Use the explicit real-model formulas, with Karatsuba-style products and a shared inversion, and verify them against the generic law.
2. **Make memory part of the comparison.** Report area-time for every Arm J row, and test a Jind-θ arm that isolates the graph from the walk.
3. **H5 follow-up.** Build a large-`p` family of Res-shaped curves with an order-4 automorphism. Measure Arm J on the automorphism-reduced graph against rho in `Jac(C)` with the same automorphism.
4. **H8.** Build a descended-`S_4` oracle over `F_{p³}` and measure the `n = 3` crossover.
5. **H7.** Enlarge the binary and odd `n = 4` cells to readable ℓ-subgroup sizes.

Erratum (2026-10-24): native work priced in GAE

This erratum was found during validation for the benchmark gate (PROTOCOL.md, Erratum V). The H2 and H6 bands price native operations at 0, 0.1 and 1 `F_p` multiplications. AGENTS.md §5a prices them at 0, 0.1 and 1 GAE.

Repriced with the same runs, the same cells and the same cost model (`validation/gae-band/`):

run	frozen band (native in M)	band with native in GAE	verdict
H2, h = 1, 28–40 bits	0.59–1.94	0.59–2.04	unresolved at every size (unchanged)
H2, h = 4, 36–40 bits	1.14–2.79	1.14–2.88	rho cheaper in every cell (unchanged)
H6, Arm J, 44 / 48 bits	0.99–2.96 / 0.77–2.32	0.99–3.04 / 0.77–2.38	unresolved (unchanged)

At the gate's three points (`I = 3M`, measured Jacobian cost, native 0, 0.1 and 1 GAE), H2 at 40 bits with `h = 1` costs 1.78×, 1.80× and 1.88× rho. That is the distance remaining to the matched rho at measured cost.

Validation (2026-10-24)

Both benchmark studies of this directory pass the validation gate (`docs/VALIDATION_GATE.md`): `VAL-20261024-wd3` (round 3) and `VAL-20261024-wd4` (round 4), `validation/validation_report.yaml`.

- **Replay.** An independent validator drew a nonce from `os.urandom` and selected 14 jobs from it. It also replayed every remaining cell.
  - All 5,228 recorded rows replay exactly in every column except wall time: round 3 has 1,368 (itemOJ and itemF), round 4 has 3,860 (H1, H2, H4 and H6).
  - The replays use the tool at d01c574a2 with `N2_DUMP`.
- **Scalars.** Every dumped answer was checked by `validation/verify_dump.rs`, which shares no code with `n2_routes.rs`.
  - It checks that ℓ is prime, `ℓP = O`, `N` is in the Hasse interval and `N·R = O`.
  - It checks that `k_found·P = Q` and `k_found = k` for every arm.
- **Recompute.** 167 headline numbers recompute from the CSVs. All 15 summary, ranking and gae-band files reproduce byte for byte.
- **Rho reference.** Negation is the strongest symmetry available: `j(E) ∉ F_p` on all 176 curves, and `j ≠ 0, 1728`. Rho and the IC run in the same process, with the same counters.
- **The gate's three points.** `I = 3M`, measured Jacobian cost, native work at 0, 0.1 and 1 GAE, 40 bits; rho is cheaper at every point.
  - Round 3, Arm J with `c = 1`: 2.79×, 2.84× and 2.98× rho.
  - H2 with `h = 1`: 1.78×, 1.80× and 1.88× rho.
- **Limitation.** The recompute independently implements the producer's definitions; it is not a blind derivation. The operation counts were shown to reproduce, not re-derived by a second implementation of the arms.

Run reports

- `reports/H1-hsweep-run-20261018T2240Z/run-report.pdf`
- `reports/H2-glue-run-20261018T2240Z/run-report.pdf`
- `reports/H4-memsweep-run-20261018T2209Z/run-report.pdf`
- `reports/H5-aut-census-run-20261018T2227Z/run-report.pdf`
- `reports/H6-large-run-20261018T2210Z/run-report.pdf`
- `reports/H7-subgroup-run-20261018T2209Z/run-report.pdf`